V&V setups, activities, and objectives for nanosatellite EPS

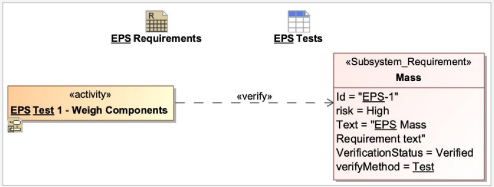
Categories of setup context and intended evidence

- Executable and analytical models. Software represents orbit, loads, generation, battery state, requirements, and failures. It supports sizing, compliance checks, and reliability analysis.
- Component bench. One physical item is measured under controlled conditions. The resulting curves supply model parameters.
- Real-time digital twin / HIL. Simulated and real components exchange signals at test speed. This exposes transients and controller behavior.
- FlatSat / EGSE. Subsystems are wired with ground equipment before full spacecraft integration. This checks interfaces, commands, and data paths.
- Environmental. A flight or qualification article faces thermal or vacuum conditions. This checks survival, battery safety, and retained function.
- Flight / telemetry. The operational spacecraft executes schedules and returns measurements. This assesses depletion risk and mission outcome.

V&V questions supported by the setups

- Desktop and analytical models. Worst-case runs, design of experiments, benchmarks, and failure analyses support sizing, compliance, and risk reduction.
- Component bench. I–V, P–V, and discharge curves support empirical solar and battery models.
- Real-time twin, CHIL, and PHIL. Physical transients and control changes support model credibility and controller robustness.
- FlatSat, EGSE, and solar simulator. Commands, data, resets, load shedding, and illumination checks support interface correctness and generation capacity.
- Thermal vacuum. Hot and cold cycles with health checks support battery safety and readiness.
- Qualification telemetry. Injected faults and residual checks support early fault isolation.
- Flight telemetry loop. Scheduled loads and measured battery response support depletion avoidance and mission yield.

SysML traceability for V&V coverage and verification status

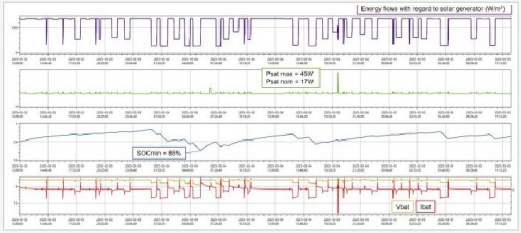




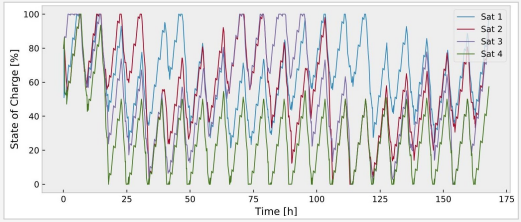
- The template links EPS test activities to requirements and records verification status.
- Its visible requirements cover power, energy storage, fault protection, load, charging, temperature, and state of charge.
- The source marks the mass example as verified and makes the remaining test links visible.
- The template supports traceability. Verified test evidence completes EPS coverage.

Source figures: *A reference architecture for rapid cubesat development*, Figures 45–46, pp. 88–89.

Mission simulations expose margin and transient depletion



48-HOUR WORST-ILLUMINATION CASE

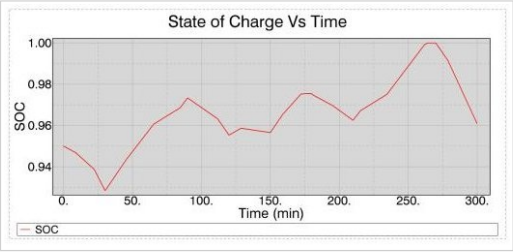


SEVEN-DAY NETSAT BATTERY STATE

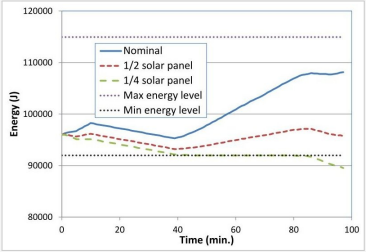
- The CNES IDM-CIC, IDM-Power, Simu-CIC, and Opalis chain varied array, battery, regulator, and architecture parameters. The selected design retained a 85% minimum state of charge.
- ESTNeT coupled orbit-control behavior, solar orientation, loads, and battery state. It found repeated depletion on one satellite that the positive average power budget missed.

Source figures: *Concurrent Engineering Process for Electrical Architecture of Nanosatellites Preliminary Phases*, Figure 10, pp. 2–6; *CubeSat energy modelling for improved mission planning and operations*, Figure 3, pp. 5–7.

Requirement-linked simulations test explicit energy margins



3DEXPERIENCE / DYMOLA STATE OF CHARGE

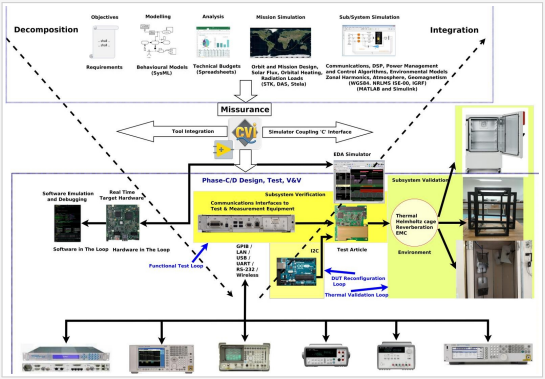


RAX SOLAR-AREA REQUIREMENT TEST

- A requirements-in-loop workflow ran hundreds of battery configurations against a 60% state-of-charge floor and recharge objective. The reported case remained above 90% for 300 minutes.
- A SysML, STK, MATLAB, and ModelCenter workflow checked a 92,000 J minimum-energy requirement. Nominal and half-size arrays passed. The quarter-size array crossed the limit.

Source figures: *An Implementation of Integrated Approach in Product Life-Cycle Management Tool to Ensure Requirements-In-Loop During Complex Product Development*, Figure 6, pp. 5–9; *Integrated model-based systems engineering applied to the Simulation of a CubeSat mission*, Figure 16, pp. 6–12.

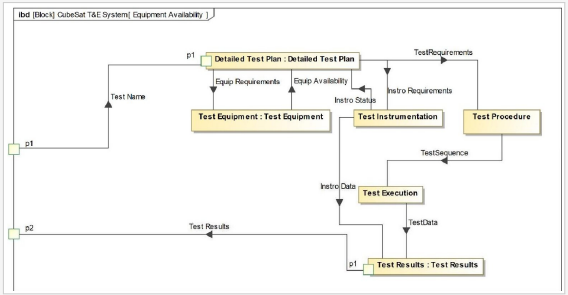
Reusable V&V platform: models, test control, and physical equipment



- Requirements, models, and budgets feed test control and instrumentation.
- Available assets include automated test equipment, HIL, the device under test, and thermal cycling.
- The electrical bench can include power supplies, a DMM, a power meter, an oscilloscope, and standard instrument links.
- The published hardware demonstration focused on communications. The platform supports future EPS testing.

Source figure: *Rapid, automated, test, verification and validation for the CubeSats*, Figure 2, pp. 7–15.

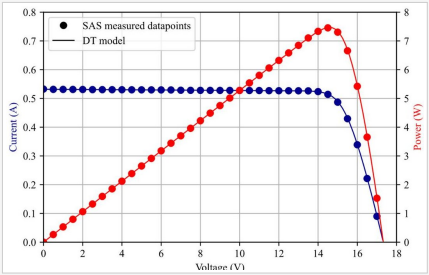
Test architecture scales from prototype to flight article



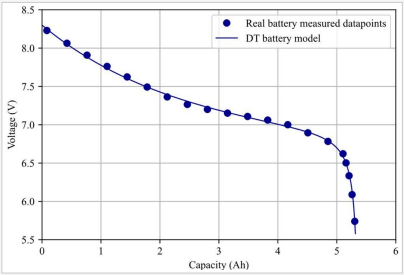
- The model connects the plan, equipment, instrumentation, procedure, execution, and results.
- EPS objectives include battery performance, solar capacity, heat, battery heaters, vibration, EMI, and telemetry power.
- Proxy procedures gave relative durations of 1.00, 1.69, and 2.78 for prototype, qualification, and flight models.
- The solar procedure used an AMSAT CubeSat Simulator and a solar energy simulator. The broader matrix remained proposed.

Source: *A Scalable System Architecture for CubeSat Test & Evaluation for Enhanced Mission Success*, Figures 25 and 47 and Appendix A, pp. 86–87, 112–113, and 142–155.

Solar and battery characterization for the digital twin



SOLAR-ARRAY CHARACTERIZATION

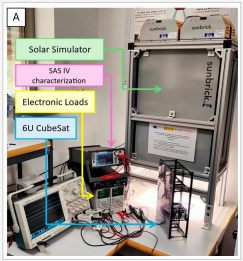


2S2P BATTERY CHARACTERIZATION

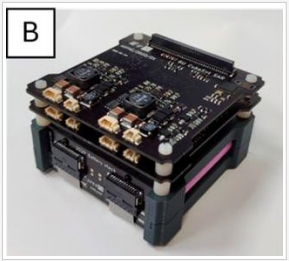
- The solar simulator measured an eight-cell solar-array section at AM0 irradiance and 85 °C. Its I–V and P–V curves parameterized the model.
- A 2S2P Samsung 18650 battery was discharged at 520 mA and 25 °C. A current step identified series resistance.

Source figures: *Implementation of a 6U CubeSat Electrical Power System Digital Twin*, Figures 2 and 7, pp. 3 and 7.

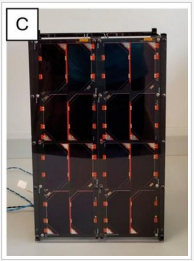
Hardware coupling to a 500 ns real-time EPS twin



HIL404 AND INSTRUMENTS



PHYSICAL EPS

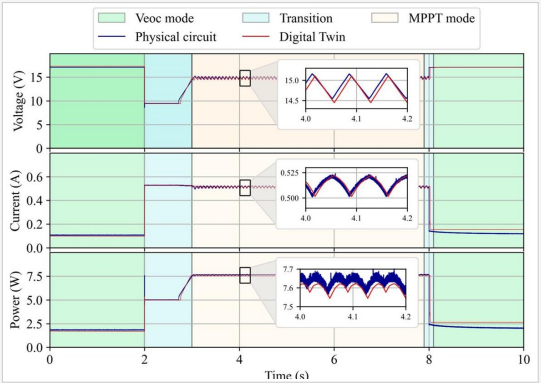


6U WITH TWO SAS

- A Typhoon HIL404 ran a twin of one solar-array section, regulator, battery, PDU, and loads with a 500 ns simulation step.
- The reported 1.65 μs loopback latency connected the physical and digital parts during MPPT, end-of-charge regulation, and a load transient.

Source figures: *Implementation of a 6U CubeSat Electrical Power System Digital Twin*, Figure 8, pp. 8–10.

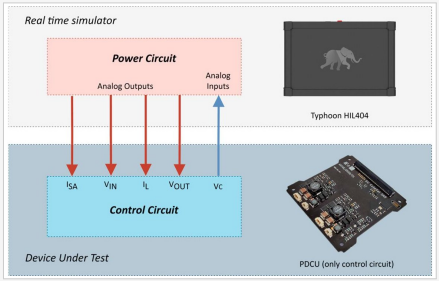
Transient EPS response to an 18 W load step



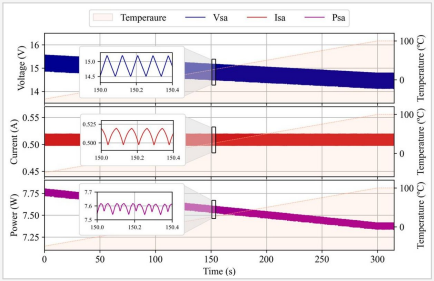
- At end of charge, solar generation exceeded load and recharged the battery.
- Adding an 18 W load at 2 s created a negative balance. The battery discharged and MPPT engaged.
- Removing the load at 8 s returned the EPS to end-of-charge regulation.
- The source reports almost identical physical and digital waveforms across this sequence.

Source figure: *Implementation of a 6U CubeSat Electrical Power System Digital Twin*, Figure 9, pp. 9–10.

CHIL MPPT tracking with a physical controller



PUBLISHED CHIL SETUP

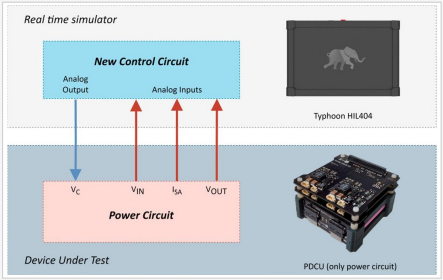


PUBLISHED CHIL RESPONSE

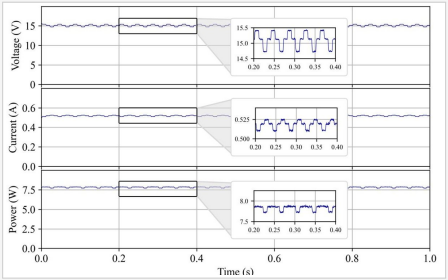
- The physical control circuit replaces the simulated regulator controller. The solar array, converter, battery, and loads remain in the real-time simulation.
- A solar-cell temperature sweep from –50 °C to +100 °C exercises MPPT tracking over expected conditions.

Source figures: *Implementation of a 6U CubeSat Electrical Power System Digital Twin*, Figures 10–11, pp. 9–11.

PHIL evaluation of a digital MPPT controller and physical power stage



PUBLISHED PHIL SETUP

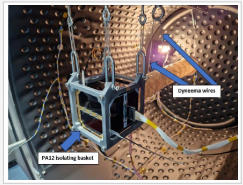


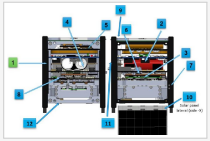
PUBLISHED PHIL RESPONSE

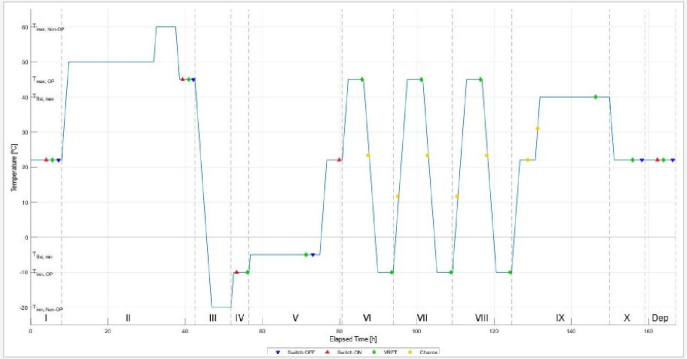
- The real-time simulator runs a digital perturb-and-observe MPPT controller while the converter, battery, and loads remain physical.
- The observed operating point oscillated around the approximately 15 V maximum-power point, supporting assessment of a control alternative.

Source figures: *Implementation of a 6U CubeSat Electrical Power System Digital Twin*, Figures 12–13, pp. 10–12.

Thermal-vacuum functional testing of the EPS flight article



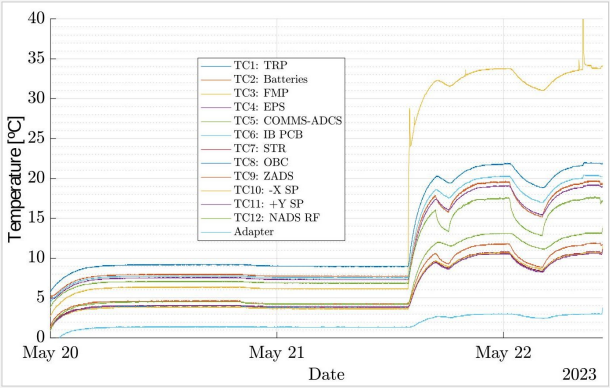




- The 3Cat-4 flight article entered a thermal-vacuum chamber with twelve thermocouples, including sensors on the batteries and EPS board.
- The campaign covered hot and cold conditions, operational and non-operational phases, charging, thermal balance, and reduced functional tests.

Source figures: *Contributions to eco-friendly satellite lean design for a sustainable space environment*, Figures 8.8, 8.10, and 8.12, pp. 203–217.

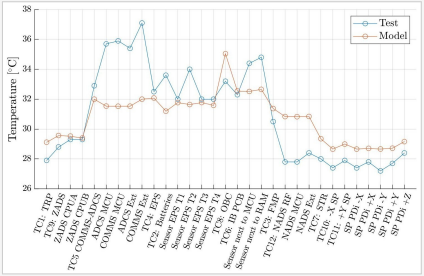
Cold battery finding, EPS configuration update, and model refinement



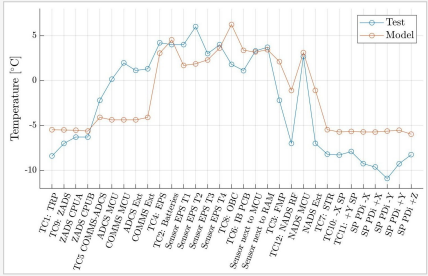
- A cold switch-on produced a Cold Batteries error. Heater activation then raised EPS and battery temperatures quickly.
- The finding triggered an EPS configuration update.
- Thermal Desktop predictions and test measurements refined conductance, material, and radiative properties.
- Pearson correlations were 0.7750 for hot balance and 0.78904 for cold balance.

Source figure: *Contributions to eco-friendly satellite lean design for a sustainable space environment*, Figure 8.15, pp. 212–219. The overall campaign met its requirements.

Hot and cold balance data correlate the thermal model



HOT BALANCE: TEST AND MODEL

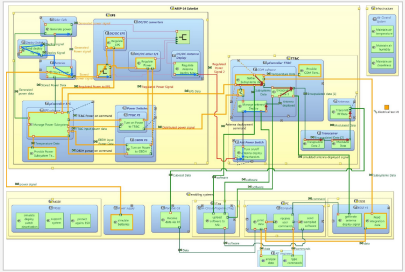


COLD BALANCE: TEST AND MODEL

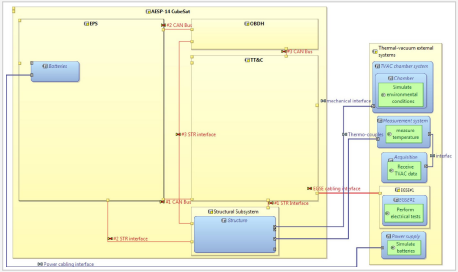
- Hot-balance correlation was 0.7750, with a maximum reported difference of 5.10 °C at the COMMS–ADCS thermocouple.
- Cold-balance correlation was 0.78904, with a maximum reported difference of 6.34 °C at the COMMS MCU sensor.
- The comparison supports refinement of interface conductance, material properties, and radiative properties.

Source figures: *Contributions to eco-friendly satellite lean design for a sustainable space environment*, Figures 8.17–8.18, pp. 217–219.

Modeled EPS functional and thermal-vacuum verification paths



BATTERY-BYPASS FUNCTIONAL PATH



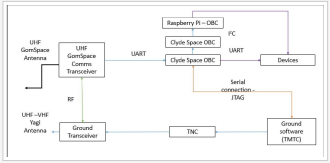
TVAC AND ELECTRICAL-TEST INTERFACES

- The AESP-14 functional path joins EPS, TT&C, a battery-bypass supply, JTAG, EGSE, a PC, a portable ground station, and mechanical support.
- The TVAC model adds the chamber, thermocouples, measurement and acquisition systems, EGSE, and power cabling that bypasses the batteries.
- Both views define functions, physical elements, interfaces, requirements, and discrepancy impact for test planning.

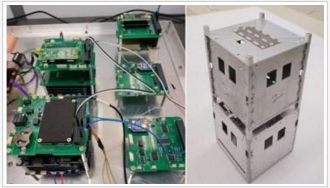
Source figures: *A conceptual MBSE framework for satellite AIT planning*, Figures 6.31 and 6.35, pp. 153–167.

DESCENT test architecture, FlatSat, and compressed sequence

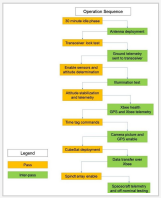
TEST ARCHITECTURE



FLATSAT



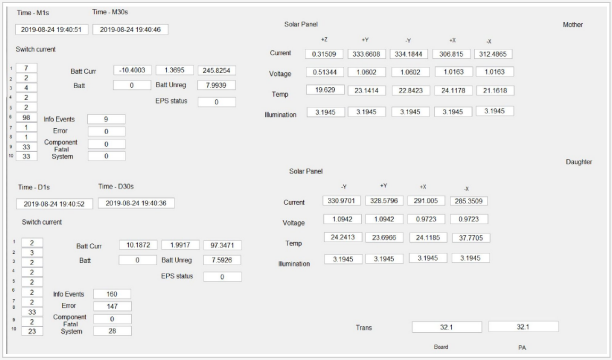
COMPRESSED TEST SEQUENCE



- Connection stages progress from direct computer commands, to a hardwired radio path, to the spacecraft and ground antenna path. The sequence compresses long operational intervals into an eight-hour campaign.
- Planned EPS checks include solar-panel illumination, resets and recovery, load shedding, and voltage, current, and temperature monitoring. The complete functional campaign remained pending.

Source figures: *Abstracting CubeSat Operations: A Path to Real Cubesat Interoperability*, Figures 23–25, pp. 109–113.

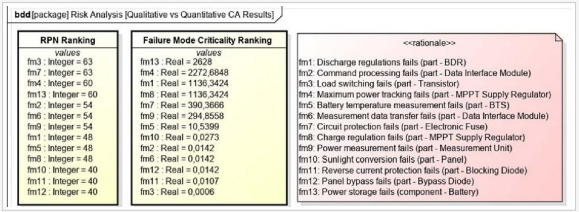
Integrated EPS telemetry during the DESCENT campaign



- The telemetry view exposes battery current and voltage, plus panel current, voltage, and temperature.
- It records switch currents and states, EPS status and resets, and events, errors, and fatal state.
- Live and delayed data support operators during a test and preserve evidence for later analysis.
- The data connect a command or load change to the resulting EPS response.

Source figure: *Abstracting CubeSat Operations: A Path to Real Cubesat Interoperability*, Figure 26, pp. 114–120.

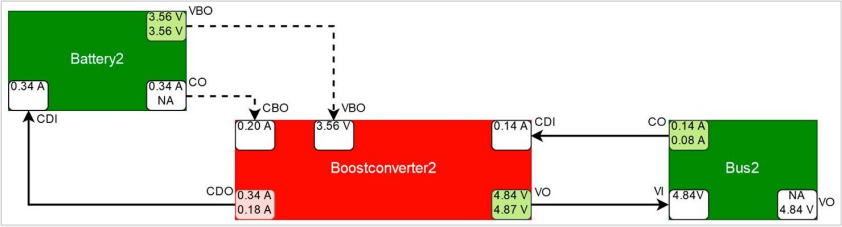
Model-based reliability analysis prioritizes EPS failure modes

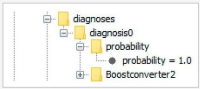


- The SysML model integrates EPS architecture with FMEA, event scenarios, criticality analysis, and fault trees.
- Thirteen modes cover regulation, switching, protection, measurement, solar conversion, reverse current, bypass, and storage.
- Quantitative ranking placed battery storage and MPPT failure highest. The qualitative and quantitative rankings differed for some modes.
- The fault tree produced 0.8802 reliability under simplifying assumptions, for comparison between designs.

Source: *Model-Based Reliability Engineering: A methodology to enhance reliability analysis in student-based CubeSat design projects*, Figures 3.12 and 3.16, pp. 57–63.

Model-based diagnosis of a boost-converter fault before a hard limit



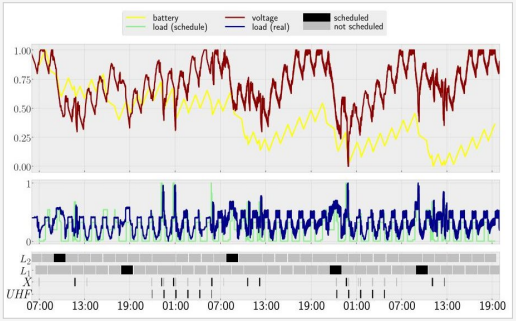


ISOLATED DIAGNOSIS

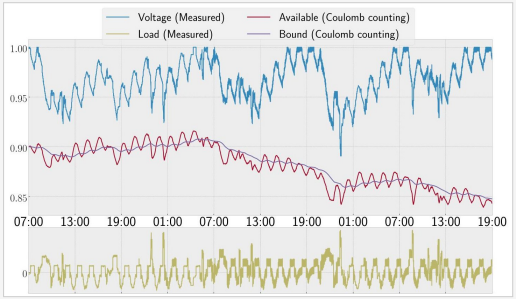
- Qualification-model housekeeping data were modified with subtle faults and compared with simulated expected values.
- In the boost-converter case, expected current was 0.18 A and measured current was 0.34 A. The diagnosis isolated Boostconverter2 before a hard telemetry limit was crossed.
- Five experiments covered single, cyclic, and indirectly observable faults. All injected discrepancies were detected and the reported root causes were identified.

Source figures and results: *Model-Based Fault Detection and Diagnosis for Spacecraft with an Application for the SONATE Triple Cube Nano-Satellite*, Figures 20 and 22 and Table 2, pp. 20–31. Faults were injected into qualification-model data.

In-orbit schedule validation with GOMX-3 battery telemetry



SCHEDULE, PREDICTION, AND TELEMETRY



MEASURED LOADS AND STATE ESTIMATES

- The 24-hour run reached 40.3% minimum state of charge, 19.3% depletion risk, and two safe-mode entries. Later 36-hour and 60-hour runs had no safe-mode entry.
- Measured current and voltage exposed drift in the battery estimate, an initial net-power mismatch, and incorrect third-party power draws.
- Telemetry closed the loop from schedule execution to model correction and later model-predictive scheduling.

Source figures and data: *Battery-aware scheduling in low orbit: the GomX–3 case*, Figures 9–10 and Table 1, pp. 13–22.

EPS sizing benchmark results: agreement and sensitivity

1U REFERENCE

Quantity	Signed error
Average power	0.00%
Orbit energy	0.66%
Battery capacity	5.20%
Effective solar area	−7.40%
True solar area	−19.30%

POWERSAT REFERENCE

Quantity	Signed error
Average power	0.00%
Battery capacity	19.02%
Battery-cell count	50.00%
Average solar power	−1.73%
Effective solar area	0.95%

- Average power matched both references. The PowerSat solar metrics differed by less than 2%.
- The 1U true solar-area result differed by −19.30%. PowerSat battery capacity differed by 19.02%, and discrete cell count differed by 50.00%.
- The benchmark supports selected calculations and exposes sensitivity to solar and battery assumptions.

Source: *Development of a Cubesat Conceptual Design Tool and Implementation of the EPS Design Module*, Tables 16–17, pp. 89–91. Values are signed percent error against two reference designs.